File Handling & Data Processing System

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Table of Contents

Introduction

Managing large amounts of data manually can be time-consuming and prone to errors. Organizations often rely on automated systems to store, process, analyze, and generate reports from their data efficiently. File handling is one of the fundamental concepts in Python that enables developers to work with files such as CSV, text, and log files.

The **File Handling & Data Processing System** is a Python-based console application developed to demonstrate practical file handling and data processing techniques using CSV files. It allows users to generate sample sales data, load records, search products, filter and sort information, generate sales summaries, and export processed data into CSV and text report files.

Unlike a basic file handling program, this project also includes automatic directory creation, activity logging, data filtering, sorting, searching, report generation, and exception handling. The application stores all data locally and requires only Python's built-in libraries, making it lightweight, portable, and easy to use without any external database or third-party packages.

This project helped me strengthen my understanding of Python file handling, CSV processing, logging, exception handling, modular programming, and basic data analysis while developing a practical real-world application.

Objectives

The main objectives of this project are:

- To develop a complete file handling system using Python.
- To read and process data from CSV files efficiently.
- To perform searching, filtering, and sorting operations.
- To generate meaningful sales summaries and reports.
- To export processed records into CSV and text files.
- To maintain application activity logs using Python logging.
- To implement robust input validation and exception handling.
- To improve programming skills using functions, modules, and file management.

Features

Sales Data Management

- Generate sample sales data automatically.
- Load sales records from CSV files.
- Display records in a structured table.
- Skip invalid or corrupted records safely.

Data Search and Filtering

- Search products by product name.
- Filter records using any available column.
- Support text-based and numeric filtering.
- Display matching records instantly.

Data Sorting

- Sort records using any column.
- Support ascending and descending order.
- Organize records for easier analysis.

Sales Analytics

- Calculate total revenue.
- Calculate total quantity sold.
- Calculate average unit price.
- Identify the highest revenue-generating product.
- Generate category-wise sales summaries.

Export Features

- Export processed records to CSV.
- Generate summary reports in text format.
- Save reports for future reference.

Logging

- Record application activities with timestamps.
- Maintain detailed log files.
- Track important file operations.

Additional Features

- User-friendly menu-driven interface.
- Automatic directory creation.
- Input validation.
- Exception handling.
- Local data storage.

Technologies Used

Technology	Purpose
Python 3	Programming Language
CSV	Data Storage
VS Code	Code Editor
Git	Version Control
GitHub	Project Repository

Python Modules Used

csv

Used for reading, writing, and processing CSV files containing sales records.

os

Used for creating directories, checking file paths, and managing file operations.

logging

Used to record application activities with timestamps for monitoring and debugging.

datetime

Used to generate timestamps for reports and log files.

statistics (mean)

Used to calculate the average unit price during sales analysis.

Project Workflow

```
Start Program
    |
    ▼
Generate or Load Sales Data
    |
    ▼
Display Main Menu
    |
    ▼
User Selects an Operation
    |
    ├── View Records
    ├── Search Records
    ├── Filter Records
    ├── Sort Records
    ├── Generate Summary
    ├── Export Data
    └── View Logs
    |
    ▼
Process Sales Data
    |
    ▼
Display Results
    |
    ▼
Export Reports (Optional)
    |
    ▼
Exit Program
```

Project Structure

```
File_Handling_System
│
├── file_handling_system.py
├── data/
│   └── sales_data.csv
├── output/
│   ├── processed_sales.csv
│   └── summary_report.txt
├── logs/
│   └── activity.log
└── README.md
```

Screenshots

```
PS C:\Users\rajes\Desktop\python\File Handling> python file_handling_system.py

==========================================================
   FILE HANDLING & DATA PROCESSING SYSTEM - SALES DATA
==========================================================
 1. Generate sample data file
 2. Load data from CSV
 3. View all records
 4. Filter records
 5. Sort records
 6. Summarize data
 7. Search records by product name
 8. Export current records to CSV
 9. Export summary report to TXT
 0. Exit
==========================================================

Enter your choice: 1
Sample data created at: C:\Users\rajes\Desktop\python\File Handling\data\sales_data.csv
```

```
==========================================================
   FILE HANDLING & DATA PROCESSING SYSTEM - SALES DATA
==========================================================
 1. Generate sample data file
 2. Load data from CSV
 3. View all records
 4. Filter records
 5. Sort records
 6. Summarize data
 7. Search records by product name
 8. Export current records to CSV
 9. Export summary report to TXT
 0. Exit
==========================================================

Enter your choice: 2
Press Enter to use default path (sales_data.csv) or type a full path:
Loaded 10 records from 'sales_data.csv'.

==========================================================
   FILE HANDLING & DATA PROCESSING SYSTEM - SALES DATA
==========================================================
 1. Generate sample data file
 2. Load data from CSV
 3. View all records
 4. Filter records
 5. Sort records
 6. Summarize data
 7. Search records by product name
 8. Export current records to CSV
 9. Export summary report to TXT
 0. Exit
==========================================================
```

```
Enter your choice: 3
ID    Product               Category    Region Qty  Price     Date
-----------------------------------------------------------------------------
P001  Wireless Mouse        Electronics South  25   599.00    2026-01-05
P002  Office Chair          Furniture   North  10   3499.00   2026-01-06
P003  Bluetooth Speaker     Electronics East   40   1299.00   2026-01-07
P004  Notebook Pack         Stationery  West   100  89.00     2026-01-08
P005  LED Desk Lamp         Furniture   South  15   799.00    2026-01-09
P006  USB-C Cable           Electronics North  60   199.00    2026-01-10
P007  Whiteboard Marker Set Stationery  East   75   149.00    2026-01-11
P008  Study Table           Furniture   West   5    4999.00   2026-01-12
P009  Mechanical Keyboard   Electronics South  20   2199.00   2026-01-13
P010  Sticky Notes          Stationery  North  150  45.00     2026-01-14

(10 of 10 record(s) shown)
```

```
==========================================================
   FILE HANDLING & DATA PROCESSING SYSTEM - SALES DATA
==========================================================
 1. Generate sample data file
 2. Load data from CSV
 3. View all records
 4. Filter records
 5. Sort records
 6. Summarize data
 7. Search records by product name
 8. Export current records to CSV
 9. Export summary report to TXT
 0. Exit
==========================================================

Enter your choice: 4
Available columns: ProductID, ProductName, Category, Region, Quantity, UnitPrice, Date
Filter by column: ProductID
Value to match: P001
Found 1 matching record(s).
ID    Product               Category    Region  Qty   Price     Date
-----------------------------------------------------------------------------
P001  Wireless Mouse        Electronics South   25    599.00    2026-01-05

(1 of 1 record(s) shown)

Set this as current working set? (y/n): Y
```

```
==========================================================
   FILE HANDLING & DATA PROCESSING SYSTEM - SALES DATA
==========================================================
 1. Generate sample data file
 2. Load data from CSV
 3. View all records
 4. Filter records
 5. Sort records
 6. Summarize data
 7. Search records by product name
 8. Export current records to CSV
 9. Export summary report to TXT
 0. Exit
==========================================================

Enter your choice: 5
Available columns: ProductID, ProductName, Category, Region, Quantity, UnitPrice, Date
Sort by column: ProductName
Descending? (y/n): y
ID    Product               Category    Region  Qty   Price     Date
-----------------------------------------------------------------------------
P001  Wireless Mouse        Electronics South   25    599.00    2026-01-05

(1 of 1 record(s) shown)
```

```
==========================================================
   FILE HANDLING & DATA PROCESSING SYSTEM - SALES DATA
==========================================================
 1. Generate sample data file
 2. Load data from CSV
 3. View all records
 4. Filter records
 5. Sort records
 6. Summarize data
 7. Search records by product name
 8. Export current records to CSV
 9. Export summary report to TXT
 0. Exit
==========================================================

Enter your choice: 6

----- SALES SUMMARY -----
Total Records      : 1
Total Quantity Sold: 25
Total Revenue      : Rs. 14975.00
Average Unit Price: Rs. 599.00
Top Product        : Wireless Mouse (Rs. 14975.00)

By Category:
  Electronics  Qty: 25     Revenue: Rs. 14975.00
--------------------------
```

```
=========================================================
   FILE HANDLING & DATA PROCESSING SYSTEM - SALES DATA
=========================================================
 1. Generate sample data file
 2. Load data from CSV
 3. View all records
 4. Filter records
 5. Sort records
 6. Summarize data
 7. Search records by product name
 8. Export current records to CSV
 9. Export summary report to TXT
 0. Exit
=========================================================

Enter your choice: 7
Enter product name keyword: Wireless Mouse
Found 1 record(s) matching 'Wireless Mouse'.
ID    Product               Category    Region  Qty   Price     Date
----------------------------------------------------------------------------
P001  Wireless Mouse        Electronics South   25    599.00    2026-01-05

(1 of 1 record(s) shown)
```

```
=========================================================
   FILE HANDLING & DATA PROCESSING SYSTEM - SALES DATA
=========================================================
 1. Generate sample data file
 2. Load data from CSV
 3. View all records
 4. Filter records
 5. Sort records
 6. Summarize data
 7. Search records by product name
 8. Export current records to CSV
 9. Export summary report to TXT
 0. Exit
=========================================================

Enter your choice: 8
Press Enter for default filename (processed_sales.csv) or type a new one:
Exported 1 record(s) to: C:\Users\rajes\Desktop\python\File Handling\output\processed_sales.csv
```

```
=========================================================
   FILE HANDLING & DATA PROCESSING SYSTEM - SALES DATA
=========================================================
 1. Generate sample data file
 2. Load data from CSV
 3. View all records
 4. Filter records
 5. Sort records
 6. Summarize data
 7. Search records by product name
 8. Export current records to CSV
 9. Export summary report to TXT
 0. Exit
=========================================================

Enter your choice: 9
Summary report saved to: C:\Users\rajes\Desktop\python\File Handling\output\summary_report.txt
```

```
=========================================================
   FILE HANDLING & DATA PROCESSING SYSTEM - SALES DATA
=========================================================
 1. Generate sample data file
 2. Load data from CSV
 3. View all records
 4. Filter records
 5. Sort records
 6. Summarize data
 7. Search records by product name
 8. Export current records to CSV
 9. Export summary report to TXT
 0. Exit
=========================================================

Enter your choice: 0
Goodbye!
```

Sample Output

```
============================================================
 FILE HANDLING & DATA PROCESSING SYSTEM - SALES DATA
============================================================

1. Generate Sample Data
2. Load Data from CSV
3. View All Records
4. Filter Records
5. Sort Records
6. Summarize Data
7. Search Records by Product Name
8. Export Current Records to CSV
9. Export Summary Report to TXT
0. Exit


Enter your choice: 6


----- SALES SUMMARY -----

Total Records       : 10
Total Quantity Sold : 500
Total Revenue       : Rs. 123456.00
Average Unit Price  : Rs. 1367.80
Top Product         : Study Table


By Category


Electronics  Qty: 145  Revenue: Rs. 87654.00
Furniture    Qty: 30   Revenue: Rs. 33990.00
Stationery   Qty: 325  Revenue: Rs. 18525.00
```

Error Handling

To make the application reliable and user-friendly, several validation and exception handling techniques have been implemented.

The system checks for:

- Missing CSV files.
- Permission errors while reading or writing files.
- Invalid or malformed CSV records.
- Incorrect numeric values.
- Invalid menu selections.
- Invalid filter or sort columns.
- File reading and writing errors.
- Missing directories (created automatically).

These validations help prevent runtime errors, improve data integrity, and provide a smooth user experience.

Future Enhancements

The project can be further improved by adding:

- Support for Excel (.xlsx) files.
- JSON data import and export.
- Record editing and deletion features.
- Graphical reports and charts.
- Multi-column sorting and advanced filtering.
- Administrator login system.
- Cloud database integration.
- Web-based version using Flask or Django.
- Graphical User Interface (GUI) using Tkinter.

Conclusion

The **File Handling & Data Processing System** was developed to gain practical experience in Python file handling by solving a real-world data management problem. Through this project, I learned how to read and write CSV files, process structured data efficiently, perform searching, filtering, sorting, and generate analytical reports.

Working on this project also improved my understanding of modular programming, logging, exception handling, and report generation. Features such as automatic sample data creation, sales analytics, exporting reports, and activity logging made the application more practical while enhancing my problem-solving abilities.

Overall, this project strengthened my confidence in building complete Python applications for file handling and data processing, providing valuable hands-on experience that will be useful in future software development projects.